

Federated Learning for TN+NTN Network Selection: Assessment and a Decentralized Research Direction

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2026-07-14

1. Purpose and scope

This note assesses federated learning (FL) as a *parallel* research direction for the TN+NTN hybrid-network project, separate from the sequential deep-RL network-selection agent already built and reported. It captures the pros and cons of applying FL to this data, the hard constraints the data imposes, and a specific decentralized direction worth pursuing. Every claim here is grounded in the actual dataset and prior discussion; where an assumption is unverified, it is flagged as such.

2. What “federated” can mean here

The learning task under this framing is a supervised one: predict the optimal radio access technology (RAT) for a user given its link KPIs. What FL changes is not the task but *where the model is trained*. Four options span a spectrum from most to least centralized:

- **Centralized (baseline)**. Pool all five datasets, train one model. The control condition.
- **Flat federated (FedAvg)**. Each client keeps its own data; a central server averages model updates. Assumes every client can reach the central server directly.
- **Hierarchical FL (HFL)**. Intermediate aggregators (naturally, the HAPS tier) sit between clients and a global model; ground and satellite clients report to a HAPS aggregator, which syncs globally over a slower path. Physically realistic for a layered network.
- **Decentralized / peer-to-peer (gossip)**. No central or intermediate server at all; nodes exchange model updates directly with reachable neighbours and aggregate locally. The most novel and the focus of Section 6.

The five datasets map cleanly onto a **Vertical Heterogeneous Network (vHetNet)** three-tier structure — terrestrial (5G, WiFi 6), aerial (UAV, HAPS), and space (LEO) — which is a standards-aligned framing to position the work against, rather than an invented category.

3. Pros — why the data suits an FL study

- **A built-in privacy/operator-boundary motivation**. Pooling raw data across a terrestrial mobile operator, a HAPS operator, and a satellite operator is unrealistic in practice for competitive and regulatory reasons. This is the standard motivating argument for FL-in-NTN, and it applies here (qualitatively).

- **Organic, not synthetically injected, non-IID data.** The tiers differ *physically*: LEO’s large Doppler swings versus near-zero terrestrial Doppler; HAPS’s mid-range propagation delay versus LEO’s much larger one; WiFi’s sub-metre-to-metre distances versus LEO’s 800-1000+ km. A model trained only on ground data genuinely struggles to generalize to LEO inputs. This is real statistical heterogeneity — exactly the regime FL aggregation methods are designed for — rather than an artificial partition.
- **Real per-tier link latency.** The data carries actual `Propagation_Delay_ms` and `Latency_ms` per link per tier. This is the strongest differentiator over typical FL-in-NTN work, which assumes abstract or textbook communication delays: genuine wall-clock training-time and communication-cost comparisons across centralized, flat, and hierarchical FL can be computed from measured delays.
- **Clean tier structure.** The vHetNet mapping gives a natural, defensible client partition and a physically meaningful aggregation hierarchy (HAPS as intermediate aggregator).
- **A natural robustness story.** A tier going offline mid-training (e.g., a LEO satellite leaving view — common in real operation) is easy to simulate, and federated/hierarchical approaches should degrade more gracefully than a centralized one. This is the core “why federate at all” argument.

4. Cons and hard constraints

- **The data is single-source synthetic.** All five datasets came from one simulation pipeline, so the federation/privacy scenario is a *constructed narrative*, not something the data itself proves. This is acceptable for a methods paper but should not be oversold.
- **UE identifiers do not truly align across files.** Four of the five files reuse the same 20 labels (UE_000-UE_019); WiFi has 7,583 distinct ones. Critically, the matching labels are **not the same physical user** — each tier was generated by an *independent* Monte-Carlo simulation, so UE_008 in the 5G file and UE_008 in the LEO file share only a string, not a shared position, time, or channel. Consequently a “one UE observed across all tiers” client design is **not scientifically valid** with this data; the only honest partition is **client = tier/RAT**.
- **That forces a small federation: five clients.** At this scale, much FL machinery collapses: client-selection methods (e.g., HiCS-FL, heterogeneity-aware selection) are meaningless because one always uses all five clients, and FedAvg-versus-FedProx differences are marginal. A flat-FL framing is therefore weak; the study has to be about *tiered/hierarchical aggregation under real link latency*, not about client selection or scale.
- **Convergence risk under strong non-IID.** With tiers this heterogeneous, aggregated training can converge slowly or to a poor compromise model; this risk grows in the decentralized setting (Section 6).
- **A label/utility function is a hard prerequisite and does not yet exist.** Every variant — centralized baseline and all FL topologies — needs a definition of “which RAT is optimal for a given KPI row.” Nothing downstream can start until this is defined.
- **Some source citations are softly sourced.** The strategy document this assessment responds to hedges at least one reference (“referenced via a listing”). Every citation must be verified independently before entering a paper.

5. Assessment of the four topologies

Topology	Realism for NTN	Novelty	Fit to this data	Verdict
Centralized	Low (ignores operator boundaries)	None (control)	Trivial	Keep as baseline only
Flat FedAvg	Low (LEO reaching a central server is a stretch)	Low	Weak (5 clients)	Baseline / control
Hierarchical FL	High (HAPS aggregator matches tiers)	Moderate, citable template	Good	Strong safe choice
Decentralized (gossip)	Highest (no central authority)	Highest	Best, if scaled per-tier	Most promising (Section 6)

6. The decentralized direction and what it lacks

Decentralized FL (DFL) — peer-to-peer gossip with no central or intermediate server — is both the most novel option and, on reflection, the *best* fit for this data. It also repairs the five-client weakness: because there is no central server dictating a flat client list, one can legitimately instantiate **multiple nodes per tier** (several ground cells, a handful of LEO satellites, a few UAVs), yielding a communication graph of tens of nodes with real spatial structure.

The central gap DFL has — and the research opportunity — is a **realistic topology**. Nearly all DFL work assumes a benign communication graph: static or slowly changing, connected, with roughly homogeneous, symmetric links. NTN violates all three assumptions:

- **Intermittent links** — a LEO peer is reachable only during a pass (minutes), then gone.
- **Graph partitions** — when satellites leave view, the graph splits into disconnected components, then heals when they return.
- **Wildly heterogeneous delay** — ground-to-ground is milliseconds; ground-to-LEO is hundreds, so a synchronous gossip round is hostage to the slowest edge.

Concrete sub-problems to work on, each reusing an asset already built in this project:

1. **Visibility-aware gossip scheduling** — exchange updates only over currently visible links and weight each exchange by live link quality (per-link RSSI/SINR are already simulated per step).
2. **Staleness- and altitude-aware asynchronous aggregation** — asynchrony is mandatory given the delays, so peers aggregate stale neighbour models; tier/altitude provides a principled staleness prior, and the weight can be derived from *measured* per-tier latency rather than assumed constants.
3. **Partition-tolerant reconciliation** — the least-studied piece: when a partition heals (a satellite re-enters view), how do the diverged component models merge back without one clobbering the other? NTN mobility produces this naturally.
4. **Convergence and evaluation without a central observer** — with no server, no node sees global accuracy, so a disagreement / consensus-distance metric across nodes is needed both to detect convergence and to evaluate the system.

The existing physics-based simulator (`noise_models/` plus the pass-geometry and visibility model

in `drl/mobility.py`) is precisely the asset that makes a *credible* NTN topology model possible — this is what most DFL-for-NTN work hand-waves with a synthetic random graph.

7. Risks, prerequisite, and recommended next step

- **Risk: it may not converge.** Strong non-IID over a sparse, partitioning graph with no global averaging step is where DFL theory is weakest, and the system may diverge. However, *characterizing when and why it fails* against a physically realistic topology is itself a defensible result, not a dead end.
- **Prerequisite (blocks everything): the label/utility function.** Define the “which RAT is optimal for a KPI row” rule as a concrete formula before any training work; it is the shared dependency for the centralized baseline and every FL topology alike.
- **Recommended next step.** Define the label function first; then build the centralized baseline and a hierarchical-FL comparison as the safe core, with decentralized (visibility-aware, partition-tolerant) gossip as the headline contribution and the simulator supplying the realistic topology.

8. Bottom line

FL is a viable parallel direction with a genuine differentiator (real per-tier latency and a physically realistic topology model). Its pros are the organic non-IID data, the operator-boundary motivation, and the measured-latency communication model; its cons are a single-source synthetic base, a forced five-client (tier-level) partition that weakens flat FL, and an unresolved prerequisite label function. The decentralized framing is the most promising because it both fixes the scale limitation and targets the one thing decentralized FL most lacks — a realistic, intermittent, partitioning, latency-heterogeneous topology — which this project’s simulator is uniquely positioned to provide.